



Water Quality Index Method for Determining Groundwater Quality Around Oil Mining Sites



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Abstract: Crude oil is a major energy resource; however, oil exploration and production activities may release petroleum hydrocarbons (PHCs), heavy metals, and other contaminants that can adversely affect groundwater quality. This study evaluates the quality of groundwater surrounding oil mining sites in Musi Banyuasin Regency, South Sumatra, Indonesia, using the Water Quality Index (WQI) method. Groundwater samples were collected from 36 sampling points, consisting of 30 study wells located near oil drilling areas and six control wells, with two control wells in each of the three sub-districts: Sanga Desa, Babat Toman, and Lawang Wetan. The analyzed parameters included pH, temperature, total dissolved solids (TDS), turbidity, color, and heavy metals, including copper (Cu), nickel (Ni), lead (Pb), and mercury (Hg). The physicochemical and heavy metal data were processed and analyzed using Python, and the WQI was calculated to determine the overall groundwater quality classification. The results indicated that all 30 study wells (100%) were classified as unfit for consumption, whereas all six control wells (100%) were classified as excellent based on the WQI classification. The primary contaminants contributing to poor groundwater quality were Ni, Pb, and Hg, which exceeded the applicable national groundwater quality standards at several sampling points. Elevated concentrations of TDS, turbidity, and color were also observed at several study wells. These findings demonstrate a substantial anthropogenic influence of oil exploration activities on groundwater quality in the study area. Continuous groundwater monitoring, improved pollution control, and appropriate environmental management by relevant authorities are therefore essential to minimize contamination and prevent further degradation of groundwater resources.

Keywords: Groundwater quality; Oil spill; Water Quality Index; Mining sites; Heavy metals; Oil exploration

1 Introduction

Crude oil is a primary energy source that plays a crucial role across various sectors of the economy. Along with the increasing global energy demand, exploration and production activities of crude oil continue to rise [1]. It is estimated that approximately 600,000 tons of crude oil are contaminated in terrestrial and aquatic ecosystems worldwide each year [2]. Crude oil exploration is conducted using various methods, including both modern and traditional techniques, with simpler approaches [3].

One of the oil-rich regions in Indonesia is Musi Banyuasin Regency, located in South Sumatra Province. Oil exploration in this area has been traditionally carried out since the Dutch colonial era and continues to this day. There are more than 500 active wells with an average production capacity of 650 barrels of oil per day (BOPD) [4]. Accidents, leaks, and spills during exploration, production, transportation, and storage pose a risk of environmental contamination, particularly to groundwater [2, 3]. The oil exploration process generates various types of waste,

including liquid waste, acid sludge, and drilling residues [4]. One of the major impacts is the deterioration of groundwater quality due to the infiltration of pollutants from exploration activities into the aquifer layer [5–7].

The most frequent type of contamination observed in the oil fields of Musi Banyuasin is oil spillage during exploration activities [4]. These activities also generate hazardous waste, such as acid sludge, which is a complex mixture of petroleum hydrocarbons (PHCs), water, heavy metals, and solid particles. More than 33% of acid sludge contains up to 550 mg/kg of polyaromatic hydrocarbons (PAHs), along with heavy metals such as barium, lead, zinc, copper, mercury, chromium, arsenic, and nickel [2, 3].

Acid sludge contamination can cause severe environmental impacts, including alterations in the chemical and physical properties of soil, changes in soil structure, a reduction in nutrient content, and inhibition of vegetation growth in affected areas [2, 3]. In addition, pollutants in acid sludge can infiltrate and contaminate groundwater sources [8]. PAHs and heavy metals are known to be genotoxic, bioaccumulative in living organisms, and may pose health risks such as neurological damage, cancer, and endocrine disruption [9]. A major challenge in addressing oil spill pollution is its persistent nature, difficulty in natural degradation, and tendency to accumulate in the environment and organisms [9, 10].

Research on oil spills' impact on groundwater is vital to assess risks. The Water Quality Index (WQI) [11] offers a comprehensive evaluation of groundwater parameters, clarifying effects of oil exploration. It also supports mitigation strategies and better environmental management to safeguard this vital resource [11, 12]. However, the WQI has several limitations: it oversimplifies the complex conditions of groundwater, is highly dependent on parameter selection, is not sensitive to local or emerging contaminants, does not indicate pollution sources, and does not account for spatial variability or data uncertainty. As a result, its assessment outcomes may be less accurate under specific conditions. Nevertheless, the WQI can be readily adapted to incorporate additional parameters without the need for complex models or sophisticated weighting schemes, making it suitable for rapid field analysis. Furthermore, it produces a single, easily interpretable value, unlike multicriteria methods that generate matrix scores, weights, or graphs that are more difficult for the public and regulators to understand.

Previous studies have examined groundwater quality from various perspectives, including groundwater characteristics, petroleum contamination, treatment technologies, and water quality assessment. Studies on the characteristics and distribution of deep groundwater have demonstrated the importance of hydrogeological conditions in determining groundwater quality and contaminant distribution [5]. In the context of oil and gas activities, previous research has also reviewed treatment technologies for produced water and emphasized the complexity of contaminants generated during petroleum exploration and production [6]. PHCs are of particular concern because they can persist in groundwater aquifers and undergo various environmental processes, including transport, biodegradation, and transformation, which may affect their toxicity and associated environmental risks [7]. In addition, the WQI has been widely applied to integrate multiple physicochemical and chemical parameters into a single indicator for evaluating groundwater suitability for drinking purposes [8]. Previous research by the author has also assessed groundwater quality in Cimahi City, West Java Province, demonstrating the applicability of water quality assessment methods for identifying groundwater conditions in areas affected by anthropogenic activities [9]. However, limited studies have specifically evaluated groundwater quality around oil mining areas in Musi Banyuasin Regency by integrating physicochemical parameters and heavy metal contamination using the WQI approach. Therefore, this study aims to assess the quality status of groundwater surrounding oil mining sites in Musi Banyuasin using the WQI method and to identify the parameters contributing to groundwater contamination.

2 Research Significance

This research is crucial, considering that groundwater serves as the primary source of water for daily life. The low service coverage of the regional water utility company in the study area has led to a high dependence of the local population on groundwater to meet their daily clean water needs. Groundwater contamination poses a significant risk to public health and can lead to economic decline, as affected communities are often forced to purchase clean water for everyday use.

3 Methodology

Groundwater samples were collected using the grab sampling method, as specified in the Indonesian National Standard (SNI) No. 6989.58:2008, which provides guidelines for groundwater sampling procedures [13]. A total of 36 groundwater samples were collected from three sub-districts in Musi Banyuasin Regency: Sanga Desa, Babat Toman, and Lawang Wetan. In each sub-district, 10 study wells located near oil drilling areas and two control wells located at sites distant from oil wells were selected, resulting in 12 sampling points per sub-district and 36 sampling points in total. Thus, the sampling design consisted of 30 study wells and six control wells, with two control wells in each sub-district. Figure 1, Table 1, Table 2, and Table 3 present the locations and characteristics of the sampling sites. In the site descriptions, “RA” refers to residential areas, “RF” refers to rice fields, and “dist.” refers to the distance between the sampling location and the nearest oil drilling well, expressed in meters.

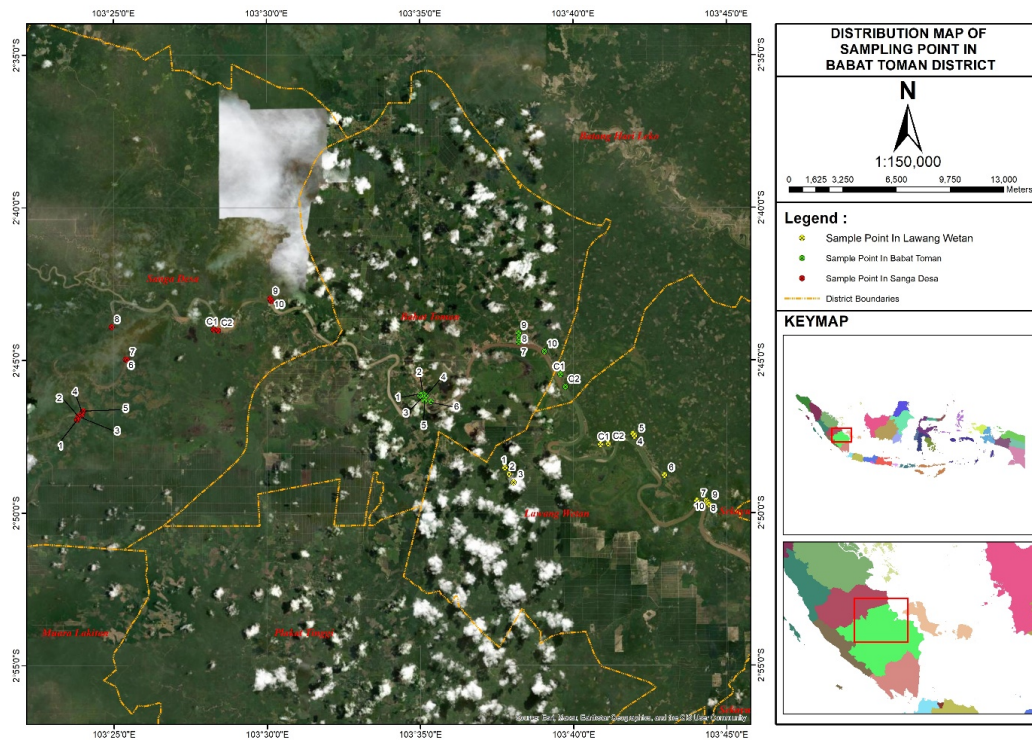


Figure 1. Sampling point location

Table 1. Sampling point location in Sanga Desa

No.	Landuse	North Coordinate	East Coordinate	Distance	Depth
1	RA	9692321.00	321774.00	0.33	2.0
2	RA	9692539.00	321863.00	0.58	3.1
3	RA	9692573.96	322033.10	0.67	2.0
4	RA	9692758.59	322085.21	0.87	2.5
5	RA	9692858.32	322159.52	1	3.7
6	RA	9695960.33	324716.20	1.1	3.2
7	RA	9695949.00	324800.00	0.1	2.0
8	RA	9697917.35	323862.75	0.17	1.9
9	RA	9699632.82	333454.08	0.33	1.0
10	RA	9699495.38	333527.81	0.27	1.0
C-1	RA	9697775.82	330037.62	4.9	0.5
C-2	RA	9697716.20	330311.82	4.6	3.0

Note: RA = residential areas.

Table 2. Sampling point location in Babat Toman

No.	Landuse	North Coordinate	East Coordinate	Distance	Depth
1	RF	9693821.10	342495.20	0.14	1.3
2	RF	9693935.62	342692.37	0.44	1.2
3	RA	9693656.11	342663.48	0.38	2.0
4	RA	9693779.49	342894.59	0.68	2.0
5	RA	9693466.50	342797.55	0.65	1.0
6	RF	9693425.57	343183.95	0.84	1.5
7	RA	9697023.76	348483.12	0.13	0.6
8	RA	9697293.66	348513.44	0.32	1.0
9	RA	9697560.70	348477.62	0.11	0.8
10	RA	9696455.84	350048.97	0.18	1.0
C-1	RF	9695072.31	351004.93	2.23	2.5
C-2	RA	9694323.33	351310.50	3.12	3.3

Note: RA = residential areas; RF = rice fields.

Table 3. Sampling point location in Lawang Wetan

No.	Landuse	North Coordinate	East Coordinate	Distance	Depth
1	RF	9689421.54	347659.95	0.53	3.2
2	RA	9689044.03	347912.13	0.46	2.6
3	RA	9688556.83	348180.53	0.69	2.5
4	RA	9691477.22	355411.66	0.2	1.0
5	RA	9691357.30	355493.99	0.33	1.0
6	RF	9688995.00	357304.90	0.73	1.0
7	RA	9687447.41	359261.37	0.42	0.5
8	RA	9687455.66	359842.73	0.56	1.0
9	RF	9687326.67	359945.70	0.63	2.7
10	RA	9687227.64	359991.58	0.66	1.0
C-1	RA	9690850.17	353442.22	6.76	2.0
C-2	RA	9690880.05	353906.18	7.26	2.5

Note: RA = residential areas; RF = rice fields.

The selection of sampling points was based on distance criteria from contamination sources [10]. Wells located within 45 meters of the contamination source were classified as having a high pollution risk, those within 200 meters as moderate risk, those within 1 km as low risk, and wells more than 1.5 km away were considered to have a very low pollution risk [10]. The main sampling points (10 in each sub-district) were selected from wells located within 1 km of contamination sources, while control points were selected from wells located more than 1.5 km away.

The groundwater quality parameters analyzed included pH, temperature, total dissolved solids (TDS), turbidity, color, and concentrations of heavy metals (Ni, Cu, Pb, and Hg). The pH analysis was conducted based on SNI No. 6989.11:2019 [14]; temperature and color based on SNI No. 06.6989.23-2005 Part 59 [15]; TDS based on SNI No. 6989.27:2019 [16]; turbidity based on SNI No. 06.6989.25:2005 [17]; and heavy metal analysis based on SNI No. 6989.69:2009 [18]. The laboratory tests were conducted at PT. Sucofindo, Mineral Division, Cibitung. The results provided a comprehensive overview of groundwater quality in the study area, including the potential impact of petroleum contamination.

The WQI method was used to determine the groundwater quality status. The WQI calculation involved selecting parameters including heavy metals (Cu, Ni, Pb, and Hg), pH, temperature, TDS, turbidity, and color [11]. The formulas used for WQI calculation are presented in the formulas below, where C_i represents the measured concentration of each parameter and S_i is the quality standard based on the Indonesian Ministry of Health Regulation No. 2 of 2023 concerning environmental health [19]. Regulation of the Minister of Health No. 2 of 2023 was selected because it is the most recent regulation that establishes environmental health-based water quality standards. It comprehensively covers physical, chemical, and biological parameters, is easy to apply for monitoring purposes, has a strong legal basis, and provides flexibility to accommodate regional needs and public health protection objectives. Each parameter was assigned a weight (w_i) according to its impact on human health and groundwater quality. The assigned w_i values are shown in Table 4.

Table 4. Weight (w_i) and Relative Weight (Wi) values [11]

Parameter	Weight (w_i)	Wi (Relative Weight) = $w_i / \sum w_i$
Pb	5	0.12
Hg	5	0.12
Ni	5	0.12
Cu	5	0.12
TDS	5	0.12
Turbidity	5	0.12
Color	5	0.12
pH	3	0.07
Temperature	3	0.07
$\sum$	41	1.00

Note: TDS = total dissolved solids.

The WQI method classifies groundwater quality into five categories: excellent (WQI < 50), good (WQI: 50–100), poor (WQI: 100–200), very poor (WQI: 200–300), and unsuitable for consumption (WQI > 300) [11]. Metals such as Ni and Pb predominantly disperse horizontally following the flow of shallow groundwater, are

constrained by impermeable vertical layers, and undergo strong sorption onto specific minerals. As a result, their concentrations tend to correlate with distance from the source rather than depth. Consequently, distance becomes the most influential factor controlling metal concentrations. The relationships between groundwater quality parameters and the characteristics of the sampling locations were evaluated based on the available physicochemical and heavy metal data. The concentrations of heavy metals, particularly Ni, Pb, and Hg, were compared with the applicable national groundwater quality standards to identify parameters that contributed to groundwater contamination. The spatial distribution of groundwater quality was also evaluated based on the sampling locations and their distances from the nearest oil drilling wells.

$$q_i = \frac{C_i}{S_i} \times 100 \quad (1)$$

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$SI = W_i \times q_i \quad (3)$$

The observed relationship between heavy metal concentrations and the distance from oil drilling wells suggests that spatial proximity to potential pollution sources may influence groundwater quality. However, the mechanisms controlling the transport and distribution of these metals cannot be determined conclusively from the present dataset and would require further hydrogeochemical and contaminant-transport investigations.

Based on their toxicity to humans and the environment, heavy metals such as Hg, Pb, Cu, and Ni exhibit different levels of hazard. Mercury (Hg) is considered the most toxic due to its strong neurotoxic properties and its ability to bioaccumulate and biomagnify within the food chain. Lead (Pb) is also highly toxic even at relatively low concentrations, as it can damage the nervous system and kidneys. Copper (Cu), although an essential trace element required for biological processes, can become toxic when present in elevated concentrations. Nickel (Ni) is generally considered less toxic compared to Hg and Pb, but prolonged exposure at high concentrations can still pose health risks and has been associated with carcinogenic effects.

Based on toxicity classification, Hg and Pb are commonly categorized as highly toxic heavy metals due to their strong bioaccumulation potential and their ability to cause long-term or permanent health damage. In contrast, Cu and Ni are generally classified as moderately to less toxic. Although these metals may be tolerated by organisms in trace amounts, their presence above permissible environmental limits can still pose significant risks to both aquatic ecosystems and human health.

In the calculation of the WQI, equal weighting is often assigned to Pb, Hg, Cu, and Ni because these metals collectively represent important indicators of heavy metal contamination in aquatic environments. Despite differences in their individual toxicity levels, all four metals are considered hazardous pollutants due to their persistence, toxicity, and potential to accumulate within aquatic organisms and sediments, which can ultimately impact human health through the food chain.

Several factors support the use of equal weighting for these heavy metals in WQI calculations. First, Hg, Pb, and Ni are generally classified as non-essential toxic metals that can cause adverse health effects even at relatively low concentrations, while Cu, although essential in trace amounts, becomes toxic when its concentration exceeds environmental quality standards. Second, these metals are known to exhibit bioaccumulation characteristics, allowing them to accumulate within aquatic organisms and sediments over time. Third, Pb and Hg are frequently used as key indicators of pollution originating from industrial and mining activities, and they are often detected together in contaminated waters. Assigning equal weights therefore ensures that the presence of any individual metal or a combination of these contaminants significantly influences the WQI value.

Furthermore, in several standard WQI calculation methods, uniform weighting is applied as a practical approach to simplify the assessment of water quality. Although some studies may assign different weights depending on specific environmental concerns—such as prioritizing mercury contamination—equal weighting is commonly adopted to represent the overall burden of heavy metal contamination. A high cumulative concentration of these metals generally indicates severe water quality degradation and potential ecological risk.

4 Results and Discussions

Musi Banyuasin Regency in South Sumatra, Indonesia, has rich petroleum potential. Oil exploration began during the Dutch colonial era, but many wells were abandoned after independence. In 1974, local communities started reusing these old wells as an alternative source of income by tapping into the available natural resources [20]. From 2012, approximately 7,734 illegal oil wells have been identified across various sub-districts in Musi Banyuasin. Uncontrolled oil extraction processes often result in oil spills that contaminate the surrounding environment, including soil and groundwater [20]. Areas within Musi Banyuasin Regency with significant petroleum potential include Babat Toman, Jirak Jaya, Sanga Desa, Keluang, Bayung Lincir, and Lawang Wetan sub-districts [21–23].

The distribution data and oil production levels served as the basis for determining sampling locations. Babat Toman Sub-district, with 500 community-managed wells and a production rate of 2,000 BOPD, was selected as a strategic research site, as its high oil output indicates a potentially greater environmental impact. The second sampling location, Sanga Desa Sub-district, has 400 community-managed wells with a production rate of 1,000 BOPD. Lawang Wetan Sub-district contains 200 wells with a production level of 800 BOPD, making it one of the regions with a relatively high oil yield.

Groundwater quality measurements at the 30 study wells for parameters including temperature, pH, TDS, turbidity, color, Cu, Ni, Pb, and Hg are presented in Figure 2, which compares these parameters across 10 sampling points in each sub-district. The six control wells were excluded from Figure 2 to provide a clearer visualization of groundwater quality variations among the study wells located near oil drilling areas. The results of the control wells are presented separately for comparison with the study wells. The quality of groundwater in the study area shows variations influenced by environmental conditions and anthropogenic activities, particularly oil exploration. The average measured groundwater temperature was 27.3 °C, which meets quality standards as it falls within a 3 °C deviation from the 2024 average ambient temperature in Musi Banyuasin Regency of 29.43 °C [20].

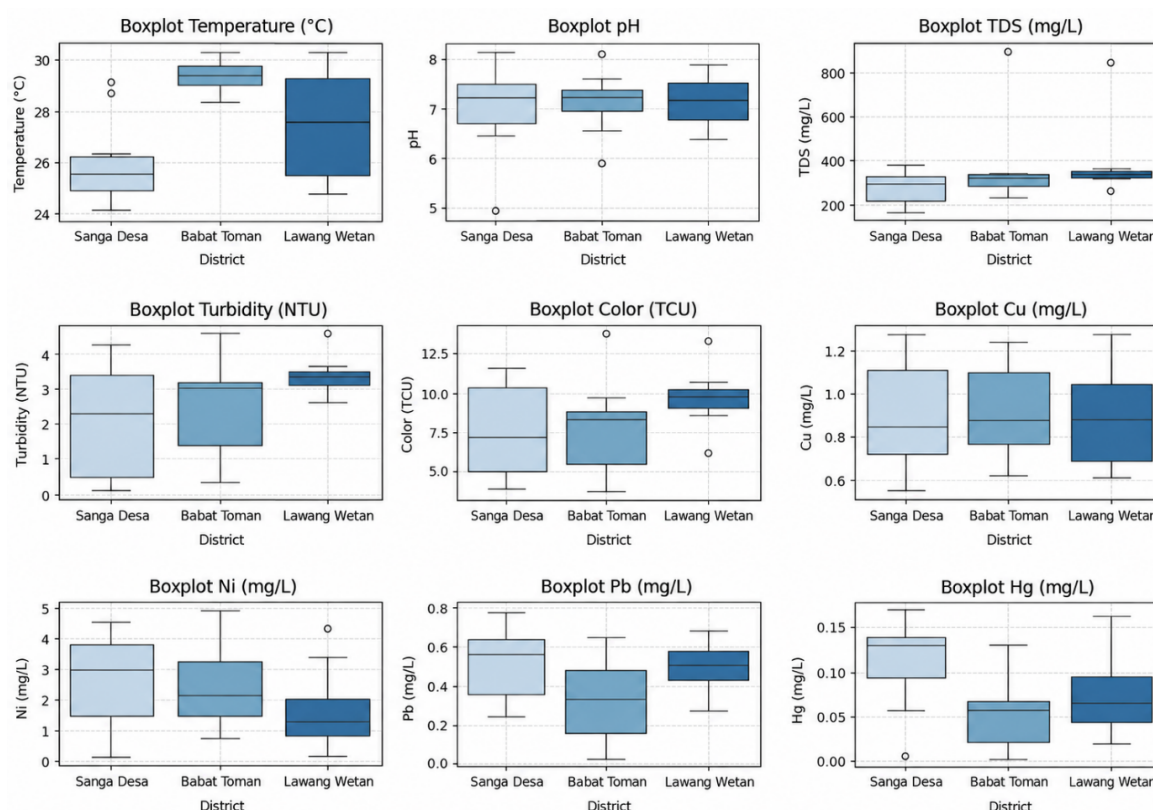


Figure 2. Groundwater quality parameters at the 30 study wells in Sanga Desa, Babat Toman, and Lawang Wetan sub-districts

Note: Standard: temperature = ± 3 deviation from air temperature, pH = 6.5–7.5, TDS = 300 mg/L, turbidity = 3 NTU, color = 10 TCU, Cu = 2 mg/L, Ni = 0.07 mg/L, Pb = 0.01 mg/L, and Hg = 0.001 mg/L (Minister of Health Regulation No. 2 of 2023, on Environmental Health).

The pH values of groundwater in Sanga Desa ranged from 4.8 to 8.2, with two sampling points (points 6 and 8) falling below the quality standard. In Babat Toman, pH ranged from 5.8 to 8.1, with one point (point 10) not meeting the standard. In Lawang Wetan, the pH ranged from 6.3 to 8.2. Low pH values in groundwater may be influenced by the aquifer's geological composition or the surrounding lithology. Acidic groundwater is commonly found in areas with alluvial deposits. Alluvium with high organic content can lower groundwater pH, especially in wetlands or swampy areas. Decomposing organic matter produces humic and fulvic acids, which contribute to groundwater acidity [20, 21].

Oil exploration activities, such as drilling, can lead to soil and groundwater contamination due to petroleum spills. Such pollution alters the physical, chemical, and biological properties of the environment, leaving toxic residues such as heavy metals that are difficult to degrade. Of the four heavy metals analyzed, Ni, Pb, and Hg exceeded permissible limits at several points in each sub-district. In contrast, the concentrations at the control wells complied with the standards. This indicates that oil extraction activities contribute to heavy metal contamination.

The groundwater quality status was assessed in the three sub-districts using parameters such as temperature, pH, TDS, turbidity, color, and heavy metals (Pb, Hg, Ni, Cu). The quality status was determined based on the concentration of each parameter. Figure 3 shows the summarized WQI values for each sub-district. The WQI calculation results indicate that groundwater contamination levels in the study area fall within the 'unfit for consumption' category. This suggests that oil drilling activities have impacted groundwater quality within a 0–1 km radius.

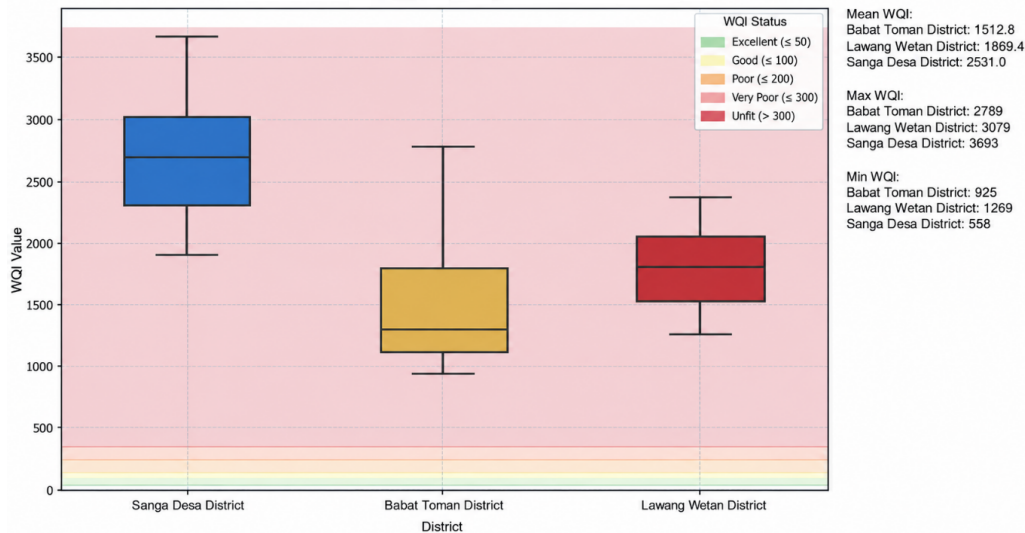


Figure 3. Water Quality Index (WQI) values for the 30 study wells located within 0–1 km of oil drilling areas

The groundwater quality status was assessed in the three sub-districts using parameters such as temperature, pH, TDS, turbidity, color, and heavy metals (Pb, Hg, Ni, and Cu). The quality status was determined based on the concentration of each parameter. Figure 3 presents the summarized WQI values for the 30 study wells, comprising 10 sampling points in each sub-district. The six control wells, which were located more than 1.5 km from the oil wells, were not included in Figure 3. The WQI calculation results indicate that the groundwater quality at all 30 study wells falls within the “unfit for consumption” category ($WQI > 300$). In contrast, the control wells were classified as “excellent” based on the WQI assessment. These results indicate that groundwater quality around the oil drilling areas was more degraded than that of the control wells located farther from the oil wells.

Groundwater contamination in the study area also exhibits a clear spatial gradient relative to the proximity of oil extraction activities. Sampling points located within a 0–1 km radius from drilling sites generally showed higher concentrations of heavy metals and poorer WQI values. In contrast, control wells located more than 1.5 km away from the oil wells consistently exhibited better groundwater quality. This pattern suggests that contamination intensity decreases with increasing distance from the pollution source, indicating that oil extraction activities function as localized point sources of groundwater pollution.

Spatial differences were also observed among the three sub-districts investigated. Babat Toman generally showed relatively higher contamination levels compared to Sanga Desa and Lawang Wetan, which may be associated with its larger number of community-managed oil wells and higher oil production capacity. Sanga Desa exhibited moderate contamination patterns, while several sampling points in Lawang Wetan showed comparatively lower concentrations of certain parameters. These inter-district contrasts highlight the influence of local operational intensity and well distribution on groundwater quality in the region.

Overall, the spatial distribution indicates that groundwater contamination is primarily localized rather than regionally widespread. Elevated contaminant levels are concentrated around areas with active or abandoned oil wells, while groundwater quality improves significantly at greater distances from these sources. This finding suggests that pollution is strongly linked to specific operational zones and has not yet spread uniformly across the broader hydrogeological system.

Statistical analysis showed very weak to moderate correlations between well depth and metal concentrations, while weak to very strong correlations were observed between the distance from the oil drilling wells and metal concentrations (Table 5). The strength of the relationships was interpreted based on the following correlation coefficient intervals: 0.000–0.199 indicates a very weak relationship, 0.200–0.399 indicates a weak relationship, 0.400–0.599 indicates a moderate relationship, 0.600–0.799 indicates a strong relationship, and 0.800–1.000 indicates a very strong relationship. Overall, the correlation results indicate that the distance between the sampling wells and

the potential pollution source had a stronger relationship with several heavy metal concentrations than well depth.

Table 5. Correlation coefficients of sampling distance and depth with heavy metal concentrations

	Distance	Depth	Cu	Ni	Pb	Hg
Distance	1.000	0.506	-0.221	-0.088	0.494	0.077
Depth	0.506	1.000	-0.117	0.101	0.421	0.328

Note: Values represent Pearson correlation coefficients.

The correlation analysis results indicate that the relationships between spatial variables and heavy metal concentrations vary in strength. Distance shows a moderate positive correlation with depth ($r = 0.506$), suggesting that sampling locations farther from the reference point tend to have greater sampling depths. Regarding heavy metal concentrations, Pb exhibits moderate positive correlations with both distance ($r = 0.494$) and depth ($r = 0.421$), indicating that Pb concentrations tend to increase with increasing distance and depth. Hg shows a weak to moderate positive correlation with depth ($r = 0.328$), while Cu and Ni display weak to very weak correlations with both distance and depth ($|r| < 0.3$). Overall, these findings indicate that the spatial distribution patterns are not uniform across the analyzed metals, with Pb showing the strongest observed correlations with the spatial variables. However, the correlation coefficients only describe the strength and direction of the observed relationships and do not establish statistical significance. Therefore, significance testing is required to determine whether the observed correlations are statistically significant and to confirm the reliability of these spatial associations.

To prevent the further spread of groundwater contamination, it is essential to implement appropriate pollution control measures, including soil and groundwater remediation techniques. The selection of a suitable remediation method must consider both technical and non-technical aspects. From a technical standpoint, priority should be given to methods that offer high efficiency, cost-effectiveness, and field applicability. Numerous technologies have been developed to remediate soil and groundwater contaminated with heavy metals, targeting five primary objectives: (1) complete degradation of pollutants, (2) extraction for treatment or disposal, (3) stabilization through immobilization and detoxification, (4) separation, and (5) recycling of clean materials.

Despite these efforts, this study has several limitations. First, groundwater quality was assessed based on a single sampling event, making it impossible to evaluate seasonal variations in heavy metal concentrations. Second, the study included 36 sampling points across three sub-districts, consisting of sampling wells in Sanga Desa, Babat Toman, and Lawang Wetan, as well as control wells. Although the sampling points covered three sub-districts, the number of samples may still limit the generalizability of the findings to other areas. Third, hydrogeological conditions were not included in the scope of analysis, as this study focused primarily on assessing heavy metal concentrations and groundwater quality near oil extraction activities. Future research should incorporate seasonal monitoring, broader sampling coverage, hydrogeological characterization, and groundwater contamination modeling to provide a more comprehensive understanding of contaminant transport and distribution in the study area.

5 Conclusions

This study assessed groundwater quality around oil mining sites in Musi Banyuasin Regency using the WQI method and heavy metal analysis. The findings demonstrate substantial degradation of groundwater quality in areas located within a 0–1 km radius of oil extraction activities. Thirty sampling points across Sanga Desa, Babat Toman, and Lawang Wetan were classified as unfit for consumption, primarily due to elevated concentrations of Ni, Pb, and Hg that exceeded national standards. In contrast, control wells consistently fell within the good to excellent categories, indicating that contamination is geographically localized.

Spatial correlation analysis revealed that the strength and direction of the relationships between metal concentrations and spatial variables varied among the analyzed metals. Pb showed the strongest association with distance, with a moderate positive correlation ($r = 0.494$), as well as a moderate positive correlation with well depth ($r = 0.421$). In comparison, Hg showed a weak to moderate correlation with depth ($r = 0.328$), while Cu and Ni exhibited weak to very weak correlations with both distance and depth ($|r| < 0.3$). Therefore, the results do not support a general conclusion that metal concentrations are strongly associated with distance from contamination sources. Instead, the spatial patterns appear to be metal-specific, with Pb showing the most pronounced observed relationship with the spatial variables. Statistical significance testing is required to determine whether these observed correlations are statistically significant and to further support interpretations of contaminant transport.

These findings indicate that oil extraction activities may contribute to localized groundwater quality degradation and may pose risks to public health, agricultural productivity, and the economic resilience of communities that rely heavily on groundwater for daily use. To mitigate further contamination, comprehensive pollution control and effective remediation measures are necessary, including technologies that enable degradation, extraction, stabilization, separation, or recycling of contaminants. However, this study is limited by its single sampling

period, restricted number of sampling points, and the absence of a hydrogeological assessment. Future studies should incorporate seasonal monitoring, expand sampling coverage, and integrate groundwater flow modeling to better understand contaminant transport dynamics and inform long-term groundwater management strategies.

Author Contributions

Conceptualization, E.W. and R.R.; methodology, E.W. and R.R.; software, E.W. and R.R.; validation, E.W., A.Z.I., R.R., M.D., and A.C.; investigation, E.W. and R.R.; resources, E.W. and R.R.; data curation, E.W., A.Z.I., and R.R.; writing—original draft preparation, E.W., A.Z.I., and R.R.; writing—review and editing, A.Z.I.; visualization, A.Z.I.; supervision, E.W.; project administration, E.W. and A.Z.I.; funding acquisition, E.W., M.D., and A.C. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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